

MASTER'S COMPREHENSIVE EXAMINATION

Part II

Applied (90 minutes)

December 9, 2006

Instructions: Answer all three questions. Open book and open notes.

1) Do statistics labs improve student performance? To study this question, a researcher took 7 sections of an intro. statistics course, and split the students in each section completely at random into two equal sized groups. The 2 groups from each section had the same teacher for lectures. Within each section, the two groups were randomly assigned, one to go to a statistics lab to work on problems, the other group were not assigned to a lab, but were given the problems to work on at home. The following are the mean scores on the final exams for the different groups.

Sections	lab group	non-lab group
1	95	75
2	77	76
3	71	73
4	79	74
5	83	79
6	87	81
7	77	74

The researcher wanted to see if there was statistically significant evidence (at the 0.05 level of significance) that using the labs will raise average scores. Carry out an appropriate test.

2) Our research hypothesis is that men are more likely to suffer complications from a particular operation, than are women. (Our null hypothesis is no difference). We gather data over time to see if there is statistically significant evidence to support our research hypothesis.

a) To study this we draw a completely random sample of 100 men and 100 women that had the operation in 1995. We find that 50 of the 100 men, and 40 of the 100 women had complications. State the null and alternative hypothesis and carry out an appropriate test at the 0.05 level of significance, and state (in terms of the application) the conclusions of your test.

b) To continue the study, we draw a random sample of 400 people who had the operation in 2004, of whom 100 were men, and 300 were women. We find that 1 of the 100 men and 6 of the 300 women have complications. When we combine the data for 1995 and 2004, we find that 51 of the 200 men, and 46 of the 400 women had complications. Using the same null and alternative hypothesis as in part (a), apply to the combined data the test at the 0.05 level of significance. State the conclusions (in terms of the application) of your test.

c) If the conclusions in part (a) and (b) are different, explain in practical terms what conclusion should be drawn from the data.